An explicit upper bound for reciprocal sums of sets with large pairwise least common multiples

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Abstract

For a positive integer n, let $\rho(n)$ be the maximum of $\sum_{a\in A} 1/a$ over sets $A \subseteq \{1,\dots,n\}$ satisfying $\operatorname{lcm}(a,b) > n$ for distinct $a, b \in A$. We prove $\rho(n) \le 1.00189053 + 4162.91/n$ for every integer $n \ge 1$. Two smaller certificates give $1.00271168 + 638.004/n$ and $1.00447967 + 133.853/n$. The proof assigns nonnegative weights to multiples, using integer cells and residue classes modulo 420, 60 and 12. Exact integer checks and an explicit Euler–Maclaurin remainder certify all constraints. As an improvement of the asymptotic leading constant, this improves the strongest bound we located in a literature search of 12 October 2026, including version 1 of this note and Chen's 1996 bound. The larger error coefficients require separate finite-n comparisons, which we give exactly. The conjecture $\rho(n) \le 1 + o(1)$ is not proved.

1 Statement and context

A set $A \subseteq \{1,\dots,n\}$ is *lcm-separated* if $\operatorname{lcm}(a,b) > n$ whenever $a, b \in A$ are distinct. Write

$$\rho(n) = \max\left\{ \sum_{a\in A} \frac{1}{a} : A \subseteq \{1,\dots,n\} \text{ is lcm-separated} \right\}.$$

The singleton $\{1\}$ is allowed, so $\rho(n) \ge 1$.

Theorem 1.1. *For every integer $n \ge 1$,*

$$\rho(n) \le 1.00189053 + \frac{4162.91}{n}, \qquad (L = 420),$$
$$\rho(n) \le 1.00271168 + \frac{638.004}{n}, \qquad (L = 60),$$
$$\rho(n) \le 1.00447967 + \frac{133.853}{n}, \qquad (L = 12).$$

The exact coefficients K of the $1/n$ terms before the upward decimal roundings are, respectively,

$$\frac{2081454577}{500000} = 4162.909154, \qquad \frac{127600737}{200000} = 638.003685, \qquad \frac{2091444}{15625} = 133.852416.$$

We prove this theorem directly from three finite arrays of nonnegative rational numbers. Every error term in the proof is explicit; no limit optimization or approximation theorem for a class of functions is used. The arrays have 39 rows, with 24, 12, and 6 columns respectively. They are supplied as JSON files, together with the verification programs.

Related work. Problem 542 in Bloom's list [1] includes the classical bound 31/30 and a separate divisor question. The conjectural strengthening $\rho(n) \le 1+o(1)$ remains distinct from those questions. The literature records $\rho(5) = 31/30$ and $\rho(11) = 4699/4620$ [2, 4]. Chen [2, Theorem 2, p. 104] proved, explicitly,

$$\rho(n) < B_C(n) := 1.0170166 + \frac{14.21}{n-58} \qquad (n \ge 59). \tag{1}$$

Weighted counting over disjoint sets of multiples already appears in that paper (Theorem 1 and Lemma 1). The weak duality principle used here is therefore prior methodology. Our contribution is the three specific periodic arrays and their certified numerical costs. Chen's later paper [3, Theorem 1] treats elements restricted to a smaller range m relative to n, and does not supply the present bound at $m = n$. Weingartner [4, Theorem 3 and Question 1] studies the version excluding 1 and gives a lower estimate; it is not an upper bound superseding (1). The note [5, Theorem 2.3, Corollary 2.6, Proposition 2.7] formulates a general occupation-weight bound and a finite acyclic flow linear program for weighted antichain problems and leaves a $1+o(1)$ certificate for this problem open [5, Question 5.1]; it supplies methodology, not a numerical bound of the present form.

Version 1 [6] proved

$$\rho(n) \le B_0(n) := 1.0087578638 + \frac{87.5686331624}{n} \qquad (n \ge 1). \tag{2}$$

The present version improves the strongest bound we located in a literature search of 12 October 2026, in the sense of a smaller asymptotic leading constant. That comparison includes our own version 1 as well as Chen (1996). The search was bounded, not exhaustive: it covered the problem page and discussion, arXiv, targeted searches, and accessible citation and formalization records. A citation index was rate-limited, another did not resolve Chen's 1996 record, and the community discussion archive was only partly searchable. No priority claim follows from this search. Section 6 states exactly where the displayed finite-n bounds improve one another.

2 Weights and their total mass

For $1 \le a \le n$ let $M_a = \{ka : 1 \le k \le n/a\}$. If A is lcm-separated then its sets M_a are pairwise disjoint: an element of $M_a \cap M_b$ would be a multiple of $\operatorname{lcm}(a,b)$ not exceeding n. Consequently, if $z_m \ge 0$ and

$$\sum_{k \le n/a} z_{ka} \ge \frac{1}{a} \qquad (1 \le a \le n), \tag{3}$$

then

$$\sum_{a \in A} \frac{1}{a} \le \sum_{a \in A} \sum_{m \in M_a} z_m \le \sum_{m=1}^{n} z_m.$$

The last inequality uses both disjointness and nonnegativity. This is weak duality, and it includes the constraint for $a = 1$.

Fix $L \in \{12, 60, 420\}$, $T = 40$, and $Q = 10^6$. Let D be the increasing list of divisors of L. The certificate provides integers $C_{j,d} \ge 0$ for $1 \le j < T$ and $d \mid L$, and we define

$$c_{j,d} = C_{j,d}/Q \quad (1 \le j < T), \qquad c_{0,d} = 0, \quad c_{j,d} = 0 \quad (j \ge T).$$

For positive integers N and $g \mid L$ put

$$F(N,g) = \sum_{k=1}^{N} \frac{c_{\lfloor N/k \rfloor,\, g\gcd(k,L/g)}}{k}, \qquad F(0,g) = 0. \tag{4}$$

We shall certify $F(N,g) \ge 1$ for every $N \ge 1$ and $g \mid L$. Set

$$z_m = \frac{c_{\lfloor n/m \rfloor,\gcd(m,L)}}{m} \qquad (1 \le m \le n). \tag{5}$$

For $N = \lfloor n/a \rfloor$ and $g = \gcd(a, L)$, writing $a = gb$ and $L = gh$ gives $\gcd(b, h) = 1$, hence

$$\gcd(ka, L) = g \gcd(kb, h) = g \gcd(k, h), \qquad \left\lfloor \frac{n}{ak} \right\rfloor = \left\lfloor \frac{N}{k} \right\rfloor.$$

For the floor identity, write $n/a = N + \theta$, $0 \le \theta < 1$, and $N = qk + r$, $0 \le r < k$; then $0 \le r + \theta < k$. Thus the left side of (3) is $F(N, g)/a$, including at $a = 1$.

For the mass estimate, consider the positive integers $k_1 < \cdots < k_s$ in $(X, Y]$ in one residue class modulo L, where $0 < X < Y$. The empty case contributes zero. Otherwise $1/k_1 \le 1/X$, and for $i \ge 2$ the spacing is L and

$$\frac{1}{k_i} \le \frac{1}{L} \int_{k_{i-1}}^{k_i} \frac{dx}{x}.$$

Summing these disjoint intervals inside $(X, Y]$ yields

$$\sum_{i=1}^{s} \frac{1}{k_i} \le \frac{1}{L} \log(Y/X) + \frac{1}{X}. \tag{6}$$

The cell $\lfloor n/m \rfloor = j$ is $(n/(j+1), n/j]$. There are exactly $\varphi(L/d)$ residues $r \in \{1, \ldots, L\}$ with $\gcd(r, L) = d$; this follows by writing $r = du$ and counting the residues coprime to L/d. Multiply (6) by the nonnegative cell coefficients and sum to obtain

$$\sum_{m} z_m \le \tau + \frac{K}{n}, \tag{7}$$

where

$$\tau = \frac{1}{L} \sum_{j=1}^{T-1} \sum_{d \mid L} \varphi(L/d) c_{j,d} \log \frac{j+1}{j}, \qquad K = \sum_{j=1}^{T-1} \sum_{d \mid L} (j+1) \varphi(L/d) c_{j,d}. \tag{8}$$

These formulas give an explicit error constant, with no asymptotic remainder.

3 The finite check

Lemma 3.1. *For every positive integer N and every $g \mid L$,*

$$F(N, g) - F(N-1, g) = \sum_{\substack{j \mid N \\ j \le T}} \big(c_{j,e} - c_{j-1,e}\big) \frac{j}{N}, \qquad e = g \gcd(N/j, L/g).$$

Proof. For $k \le N - 1$, the floor $\lfloor N/k \rfloor$ increases precisely when $k \mid N$, and then by one. The new term $k = N$ is covered by $c_{1,g} - c_{0,g}$. Substitute $j = N/k$. Terms with $j > T$ vanish, but the term $j = T$ must be retained because $c_{T-1,e}$ need not vanish. At $N = 1$ the identity reads $F(1, g) = c_{1,g}$. □

Put $S = 2^{64}$. For every increment in Lemma 3.1, the program `check.c` adds

$$\left\lfloor \frac{(C_{j,e} - C_{j-1,e}) jS}{N} \right\rfloor \quad \text{to } \mathrm{lo}_g, \qquad \left\lceil \frac{(C_{j,e} - C_{j-1,e}) jS}{N} \right\rceil \quad \text{to } \mathrm{hi}_g.$$

Starting at zero, addition of these directed bounds preserves

$$\mathrm{lo}_g \le QSF(N, g) \le \mathrm{hi}_g \tag{9}$$

at every step. Signed integer division implements the floor and ceiling also for negative numerators. The test $\mathrm{lo}_g \ge QS$ proves feasibility; $\mathrm{hi}_g < QS$ is a violation, and $\mathrm{lo}_g < QS \le \mathrm{hi}_g$ is ambiguous and does not certify it.

For these arrays, $0 \le C_{j,d} \le 1500031 < 2^{21}$ and $T = 40 < 2^6$. The numerator has absolute value below 2^{91}. For $N \le 10^7 < 2^{24}$, partitioning $1, \ldots, N$ into $\{1\}$ and at most 24 dyadic blocks gives $\sum_{k \le N} 1/k \le 25$. At an intermediate update, each term has either its old or its new nonnegative coefficient, so the exact

scaled sum is at most $25 \cdot 1500031S$. There have been at most $40N$ rounded increments. Therefore every accumulator has absolute value less than

$$25 \cdot 1500031 \cdot 2^{64} + 400000000 < 2^{90}.$$

The signed 128-bit calculation cannot overflow. The ordinary integer indices also fit on a platform with 64-bit `long`.

The exact checks gave zero violations and zero ambiguous cases for all $g \mid L$. The original verification and two independent reviews checked $1 \le N \le 10^7$, a total of 420000000 pairs (N, g) across the three certificates. The independent Node program deposited with this version was run for $1 \le N \le 100000$, checking 4200000 pairs. This shorter range already suffices with the tails below. The smallest normalized lower margin for $L = 12$ and 60 is $4/10^6$, at $N = 1$; for $L = 420$ the minimum occurs at $(N, g) = (2586, 1)$ and its numerator in units of QS is 234328714177376032570. All are strictly positive. The Node program additionally compared (9) against direct sum enclosures at $N = 1, 2, 5, 11, 40, 41, 997, 2586$ for every g, a total of 336 checks. These checks also prove the direct lower enclosures at those points are at least QS.

4 An explicit tail estimate

Let $\widetilde{B}_1(u) = \{u\} - 1/2$ and $\widetilde{B}_2(u) = \{u\}^2 - \{u\} + 1/6$. Then $|\widetilde{B}_1| \le 1/2$, $|\widetilde{B}_2| \le 1/6$, and $\widetilde{B}_2' = 2\widetilde{B}_1$ away from integers; $\widetilde{B}_2$ is continuous at integers. For fixed g put $e(r) = g \gcd(r, L/g)$ and define

$$F_\infty(g) = \frac{1}{L} \sum_{j=1}^{T-1} \sum_{r=1}^{L} c_{j,e(r)} \log \frac{j+1}{j},$$

$$h_j(x) = \sum_{r=1}^{L} (c_{j-1,e(r)} - c_{j,e(r)}) \widetilde{B}_1\left(\frac{x-r}{L}\right),$$

$$\Psi_g = \sum_{j=1}^{T} j \sup_x |h_j(x)|, \qquad R_g = \sum_{j=1}^{T-1} \sum_{r=1}^{L} c_{j,e(r)} \frac{L(j+1)^2}{6}.$$

In particular $F_\infty(1) = \tau$.

Lemma 4.1. *For every positive integer N and every $g \mid L$,*

$$F(N, g) \ge F_\infty(g) - \frac{\Psi_g}{N} - \frac{R_g}{N^2}. \tag{10}$$

Proof. Fix $j < T$ and $r \in \{1, \dots, L\}$, and put $X = N/(j+1)$, $Y = N/j$, $a = (X - r)/L$, $b = (Y - r)/L$, $f(u) = 1/(Lu + r)$. Since $a > -r/L \ge -1$, the integers $i \in (a, b]$ correspond exactly to the positive integers $k = Li + r \in (X, Y]$. The pole of f lies strictly to the left of a. Integration by parts on the intervals between consecutive integers, using the downward jumps of size 1 of $\widetilde{B}_1$, gives

$$\sum_{a < i \le b} f(i) = \int_a^b f(u)\, du + \widetilde{B}_1(a) f(a) - \widetilde{B}_1(b) f(b) + \int_a^b \widetilde{B}_1(u) f'(u)\, du.$$

The convention $\widetilde{B}_1(\text{integer}) = -1/2$ includes b and excludes a if either is integral. Here the integral is $L^{-1} \log(Y/X)$, $f(a) = 1/X$, and $f(b) = 1/Y$. A second integration by parts gives the following bound for the final integral E:

$$|E| \le \frac{|f'(a)| + |f'(b)|}{12} + \frac{1}{12} \int_a^b |f''(u)|\, du$$
$$= \frac{|f'(a)|}{6} = \frac{L}{6X^2}.$$

Indeed $f'(u) = -L/(Lu + r)^2$ and $f''(u) = 2L^2/(Lu + r)^3 > 0$, so the last integral is $|f'(a)| - |f'(b)|$.

Multiply by $c_{j,e(r)} \geq 0$ and sum over cells and residues. The main terms give $F_\infty(g)$ and the absolute remainders sum to at most R_g/N^2. The boundary contribution is

$$\frac{1}{N}\sum_{j=1}^{T-1}\sum_{r=1}^{L} c_{j,e(r)}\left[(j+1)\widetilde{B}_1\left(\frac{N/(j+1)-r}{L}\right) - j\widetilde{B}_1\left(\frac{N/j-r}{L}\right)\right].$$

Shifting the index in the positive part, using $c_{0,d} = c_{T,d} = 0$, turns this into $N^{-1}\sum_{j=1}^{T} jh_j(N/j)$. Its absolute value is at most Ψ_g/N, proving (10). □

Exact evaluation. Each h_j is L-periodic and affine on $(x_0, x_0{+}1)$ for integral x_0, with slope $L^{-1}\sum_r(c_{j-1,e(r)} - c_{j,e(r)})$. Its supremum in absolute value is the maximum over its value at x_0 and its left limit at x_0+1, for $x_0 = 0, \ldots, L-1$. Thus Ψ_g and R_g are rational and computed exactly. The NumPy integer matrix calculation in `tail2.py` is safe: its entries, including the slope correction, have absolute value at most $(L^2+2L)1500031 < 2^{39}$, within signed 64-bit arithmetic. The independent Node calculation instead uses a recurrence. With $\delta_r = C_{j-1,e(r)} - C_{j,e(r)}$ and $H(x) = 2LQh_j(x)$,

$$H(x+1) = H(x) + 2\sum_r \delta_r - 2L\delta_{x+1}.$$

It checks both endpoints and periodic closure exactly with integers.

For logarithms, put $y = 1/(2j+1)$ and

$$\ell_j = 2\sum_{i=0}^{39}\frac{y^{2i+1}}{2i+1}, \qquad u_j = \ell_j + \frac{2y^{81}}{81(1-y^2)}.$$

Integrating the geometric series for $1/(1-y^2)$ shows $\ell_j \leq \log((j+1)/j) \leq u_j$: every omitted positive term has denominator at least 81. For $j \geq 1$ the width is at most $1/(36\cdot 3^{81})$. The sum of coefficients in any F_∞ or τ calculation is at most $39 \cdot 1.500031 < 59$, so the total logarithm enclosure has width below $59/(36\cdot 3^{81})$. Write $F_\infty^-(g)$ for the expression using ℓ_j, and τ^+ for (8) using u_j. All comparisons of these rational numbers are exact.

5 Certificate values and conclusion

Table 1 gives exact rational data at $g = 1$; its column m satisfies $F_\infty^-(1) - 1 \geq m$. In this order the exact

L	m	Ψ_1	R_1	N_0
12	$4479665243/10^{12}$	16566037/750000	1188294863/250000	5185
60	$2711678024/10^{12}$	2254215137/60000000	10493256427/100000	16255
420	$1890528110/10^{12}$	3518369257/52500000	232154535249/50000	70483

Table 1: Exact tail constants and retained sufficient onsets.

values of $mN_0^2 - \Psi_1 N_0 - R_1$ are

$$\frac{138313568659001}{120000000000}, \qquad \frac{12842678216509}{15000000000}, \qquad \frac{7574273319816137}{300000000000},$$

all positive. The corrected $L = 420$ values are used throughout. For every divisor $g > 1$ simultaneously, Table 2 gives $F_\infty^-(g) - 1 \geq m_*$, $\Psi_g \leq P_*$ and $R_g \leq R_*$. These bounds check the errors as well as the limiting margin. Because $\Psi_g, R_g \geq 0$, the expression $F_\infty^-(g) - 1 - \Psi_g/N - R_g/N^2$ is nondecreasing for $N > 0$. The positive comparisons at N_0 therefore prove $F(N,g) \geq 1$ for all $N \geq N_0$ and all g. Each $N_0 < 100000$, so this overlaps the finite check of Section 3. There is no uncovered positive integer N.

The objective calculation gives the upward-rounded rational decimals in Table 3; the exact K values appear in the theorem. The gaps between the headline leading constants and the displayed bounds for τ^+ are respectively 0.000000001889389, 0.000000001975031, and 0.000000004756242. The headline $1/n$ coefficients exceed K by 0.000846, 0.000315, and 0.000584. These are positive exact decimals. For each integer $n \geq 1$, (5) now satisfies every constraint (3), and (7) proves Theorem 1.1.

L	m_*	P_*	R_*	$10^{12}(m_*N_0^2 - P_*N_0 - R_*)$
12	$33442881402/10^{12}$	17	5473	805467948259683450
60	$21435519688/10^{12}$	26	118639	5122531725449792200
420	$15172074849/10^{12}$	49	5068429	66850545939558828361

Table 2: Conservative exact bounds for all $g > 1$.

L	$\tau \le \tau^+ \le$	exact K (decimal)	headline bound
420	1.001890528110611	4162.909154	$1.00189053 + 4162.91/n$
60	1.002711678024969	638.003685	$1.00271168 + 638.004/n$
12	1.004479665243758	133.852416	$1.00447967 + 133.853/n$

Table 3: Every displayed upper bound is rounded upward.

6 Exact comparisons with earlier bounds

Remark 6.1 (the bound from version 1). Version 1 proves $\rho(n) \le B_0(n) = 1.0087578638 + 87.5686331624/n$ for every integer $n \ge 1$. Denote the three displayed theorem bounds by B_{12}, B_{60}, B_{420}. Table 4 states all pairwise comparisons of these four functions. In each row, the bound in the second column is strictly smaller precisely for integers n at least the final-column value; it is strictly larger at every smaller positive integer. There is no integer equality.

old	new	exact real crossing	first integer with new $<$ old
B_0	B_{12}	77140611396/7130323	10819
B_0	B_{60}	2752176834188/30230919	91039
B_0	B_{420}	20376706834188/34336669	593439
B_{12}	B_{60}	50415100000/176799	285155
B_{12}	B_{420}	201452850000/129457	1556138
B_{60}	B_{420}	6408920000/1493	4292646

Table 4: Exact comparisons of the published decimal bounds.

To justify the table, write $B_i(n) = a_i + b_i/n$. In every row $a_{\text{old}} > a_{\text{new}}$ and $b_{\text{new}} > b_{\text{old}}$, and

$$B_{\text{new}}(n) < B_{\text{old}}(n) \quad \Longleftrightarrow \quad n > \frac{b_{\text{new}} - b_{\text{old}}}{a_{\text{old}} - a_{\text{new}}}.$$

The displayed reduced rational crossings and their integer floors were computed exactly. Thus the smallest of these four bounds is B_0 for $1 \le n \le 10818$, B_{12} for $10819 \le n \le 285154$, B_{60} for $285155 \le n \le 4292645$, and B_{420} for $n \ge 4292646$. These comparisons concern the stated decimal inequalities, not any unpublished sharpening of their constants.

For completeness Table 5 compares each with Chen's explicit function B_C on its domain $n \ge 59$. Outside the intervals listed, the bound in the first column is strictly larger than B_C; there is no integer equality. Indeed $n(n-58)(B_i(n) - B_C(n))$ is the concave quadratic

$$(a_i - 1.0170166)n(n-58) + b_i(n-58) - 14.21n.$$

Exact opposite signs at the adjacent endpoints in the table locate its two real roots; for B_{420} the lower root lies in $(58, 59)$. A concave quadratic is positive between its roots and negative outside. The accompanying audit records all endpoint signs as exact fractions. This proves every stated comparison.

Remark 6.2 (small values). The exhaustive searches reported with version 1 found $\rho(5) = 31/30$, $\rho(11) = 4699/4620$, and $\rho(n) = 1$ for every other $n \le 29$. This remains an observation from that version, not a theorem or a new computation in this version; its search programs are not deposited. None of the present proof depends on it. The values at 5 and 11 also occur in the literature cited above.

bound	integers $n \geq 59$ for which it is strictly below $B_C(n)$
B_0	$59 \leq n \leq 69$ or $n \geq 8872$
B_{12}	$59 \leq n \leq 64$ or $n \geq 9537$
B_{60}	$n = 59$ or $n \geq 43606$
B_{420}	$n \geq 274275$

Table 5: Comparison with Chen (1996).

7 Reproduction and verification status

The certificate files are `code/v2/cert_L420.json`, `cert_L60.json`, and `cert_L12.json`. Their fields `L`, `T0`, `Q`, `D`, `C` mean L, $T = 40$, $Q = 10^6$, the ordered divisor list, and `C[j-1][i]`$= C_{j,D_i}$ (zero-based column index i). Their respective numbers of nonzero entries are 282, 167, and 99. All other entries are explicit zeros. The certificate arrays are unchanged from those reviewed.

Run `sh code/v2/run.sh` from the bundle directory. It compiles `check.c`, then invokes `verify.py` for each certificate with $N_1 = 10^7$. Requirements are a C compiler supporting signed `__int128`, a platform with 64-bit `long`, Python 3, and NumPy. No optimizer is required. The tail calculation uses exact rational logarithm bounds and integer arrays. Its floating-point fields and the C minimum-margin displays are supplementary diagnostic images only. The final verified inequality uses upward decimal rounding of the exact rational values of τ^+ and K. The expected-output notes distinguish predictions for that run from outputs actually produced in this preparation environment.

An independent exact calculation is supplied in `audit.mjs`. Run `node code/v2/audit.mjs`; its default finite limit is 100000. The saved `audit-output.json` records that run, the exact tail values, directed objective decimals, pairwise bound comparisons, and resource use. This is an independent implementation, not an execution of the C/Python files. The preparation environment had Node but no C compiler, Python, or TeX installation. A later independent check compiled `check.c` and ran `verify.py` on all three certificates up to $N = 10^7$; its output equals the block in `code/v2/expected-output.md`. The supplied `main-v1.pdf` is the archived nine-page version 1, not this manuscript. The version 1 programs and certificate remain unchanged in `code/v1/`.

The certificate proof received two independent reviews with the verdict PASS-WITH-REPAIRS and a separate exact verifier check. The reviews requested documentary repairs: correcting the two $L = 420$ tail entries while retaining onset 70483; restoring four C and five Python string escapes damaged in source storage; and printing certified bounds with upward rounding. The two independently repaired source copies agree byte for byte. The deposited C and tail sources are those copies. The Python wrapper additionally has the output-only change recorded in `printing.patch`, and an internal provenance tag was removed from a comment. The feasibility tests, certificate arrays, tail inequalities, and verification decision are unchanged. The original verifier check used the same C source with an alternative C compiler; the independent reviews also replayed the full finite range in exact integer implementations. The present Node run provides a further, shorter exact check with complete tail overlap. No formal proof-assistant verification is claimed.

The arrays were found by floating-point linear-programming searches followed by rational rounding and feasibility checks. These searches and any floating-point sanity checks are supplementary; the proof is the exact certificate verification described above. The periodic-dual method suggests limit linear programs as a possible subject for further study.

References

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